
The Compression Hypothesis: Uncovering Predictive Signal in LLM Forecasts

Anonymous Author(s)

Affiliation

Address

email

Abstract

Large language models (LLMs) are increasingly evaluated as forecasters of real-world events with mixed results: certain systems can match expert humans on certain benchmarks, while off-the-shelf LLM forecasts continue to underperform prediction-market consensus. We seek to understand this gap. Studying 739 live, out-of-sample binary contracts from Polymarket, we find that LLM forecasts underperform market prices on standard pooled metrics, but that this underperformance has a specific structure: forecasts are systematically compressed toward 0.5, with non-uniform magnitude. We articulate the *confidence-weighted compression hypothesis*: LLMs hedge in proportion to their self-reported confidence, so high-confidence forecasts approach calibration while low-confidence forecasts retreat toward indifference. We formalize this as a confidence-dependent shrinkage of latent beliefs and derive a sharp prediction: compression damages forecast quality only when the task requires placing precise probability mass on a narrow event; tasks rewarding directional judgment over a broad outcome region preserve signal. Partitioning contracts along this axis confirms the prediction. The LLM’s edge over the market flips sign on the directional subsample (accuracy +0.049 vs. −0.026 on narrow contracts; Brier improvement +0.021 vs. −0.011), and the gap is economically substantial: a payoff edge of roughly \$0.08 per dollar staked and a +30.7% ROI on wide-region contracts versus −3.8% on narrow-region contracts.

1 Introduction

Large language models (LLMs) are increasingly evaluated as general-purpose forecasters of real-world events, spanning electoral outcomes, financial markets, sporting events, and geopolitical developments. Results have been mixed, ranging from human-level performance in some evaluations [Halawi et al., 2024, Alur et al., 2025] to clear underperformance against market and expert baselines in others [Karger et al., 2025].

We seek to understand this gap. Aggregate performance metrics can mask substantial heterogeneity in where and how models fail. If errors concentrate on identifiable subclasses of questions, metrics such as Brier score will obscure regions over which LLMs carry informative signal. The practical question — when can an LLM forecast be trusted? — requires understanding this structure.

To investigate, we collect 739 live, out-of-sample binary contracts from Polymarket spanning finance, sports, social-media engagement, and current events. For each contract, we elicit forecasts from frontier LLMs and record contemporaneous market prices. Consistent with prior work, LLM forecasts underperform market baselines on standard pooled forecast-quality metrics. However, this underperformance has structure: we find that forecasts are systematically compressed toward 0.5, and that the magnitude of compression is not uniform across questions.

Motivated by this finding, we articulate the *confidence-weighted compression hypothesis*: LLMs modulate how much they hedge based on their own self-reported confidence, so that high-confidence forecasts approach calibration while low-confidence forecasts retreat toward indifference. We formalize this as a confidence-dependent shrinkage of a latent belief toward the uniform prior,

$$p_{\text{report}} = \lambda(c) p_{\text{latent}} + (1 - \lambda(c)) \cdot \frac{1}{2}, \quad (1)$$

with mixing weight $\lambda(c) \in [0, 1]$ controlled by stated confidence c . This formulation leads to a testable prediction: compression should damage forecast quality only when the task requires placing precise probability mass on a narrow event. When a task instead rewards directional judgment over a broad outcome region, hedging preserves rather than destroys predictive signal.

We test this prediction by partitioning contracts along a wide-versus-narrow axis defined *ex ante* from the contract structure. Although every contract is binary, the underlying forecasting task differs. Wide contracts ask whether an outcome falls on a given side of a threshold (whether a stock closes above a strike, whether a candidate wins a state, whether a team covers a spread), so a forecaster who is roughly right about direction gets the call right even if its probability is hedged toward 0.5. Narrow contracts ask whether an outcome falls in a tight interval—say, the exact bucket a CPI print falls into, or whether a tweet count lands in a given range—so a forecaster needs to place its probability near the true (typically low) value, and hedging toward 0.5 directly inflates the forecast above the truth. A third category, event-style contracts (“Will France win the match?”; “Will the President say ‘crypto’ during the address?”), asks about named events with no ordered outcome space and benefits from neither property.

The split inverts the headline result. On the directional subsample, the LLM’s edge over the market flips sign: +0.049 on accuracy (versus −0.026 on the narrow subsample) and +0.021 on Brier improvement (versus −0.011). The effect is also economically substantial. Restricted to contracts the hypothesis identifies as favorable, payoff-weighted edges reach roughly \$0.08 per dollar staked, with an ROI of +30.7% (versus −3.8% on narrow contracts).

Contributions. Our work makes three contributions.

- **Empirical structure of LLM forecast errors.** Using 739 live Polymarket contracts, we document that LLM forecasts exhibit systematic, non-uniform compression toward 0.5—a structural feature that pooled forecast-quality metrics fail to surface.
- **A formal hypothesis with empirical validation.** We articulate and formalize the *confidence-weighted compression hypothesis* as a confidence-dependent shrinkage of latent beliefs (Eq. 1), and derive its predictions about which tasks should preserve and which should destroy LLM signal. We confirm the central prediction: on directional contracts, LLMs outperform market baselines whereas on narrow contracts they underperform.
- **A practical criterion.** We identify a concrete, ex-ante criterion under which LLM forecasts carry economically meaningful information, reframing the question from “do LLMs forecast well?” to “which forecasting questions are LLMs well-suited for?”

2 Revisiting the standard LLM forecasting evaluation

This section introduces the contract sample, elicitation protocol, and evaluation metrics used throughout the paper, and reports our headline pooled result against market prices. The same setup is reused in Section 3 to test the confidence-weighted compression hypothesis.

2.1 Experimental setup

Sample. Our analysis sample consists of $N = 739$ live, out-of-sample binary contracts from Polymarket,¹ collected on a rolling basis over a five-day window in March 2026 and forecast before resolution. The median time-to-resolution is approximately two days, so no look-ahead is possible: the model forecasts unresolved contracts, and performance is computed only after they resolve. We deliberately span several domains so that results are not tied to a single forecasting context: financials (44.1%), sports (32.2%), politics (22.7%), and a small economics subsample (0.9%); see Appendix A for the full breakdown with example contracts. Table 1 reports the breakdown across contract types.

¹Accessed via <https://docs.polymarket.com/api-reference/introduction>.

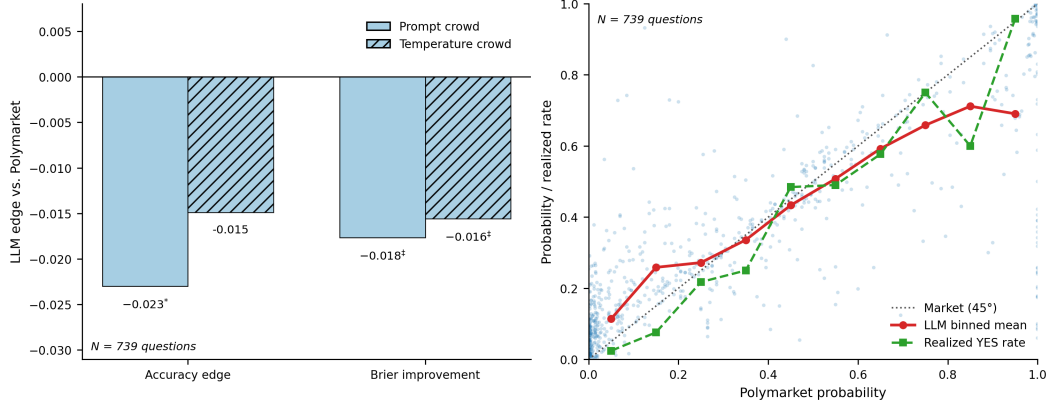


Figure 1: LLMs have a negative edge against the market (left) because their probabilities are compressed toward 0.5 (right). Left: mean LLM edge over the market on accuracy and Brier improvement ($N = 739$ questions). Right: each dot is one question; the red curve is the binned-mean LLM probability, green is the realized YES rate (the fraction of contracts in each bin that actually resolved YES), and the dotted line is the 45-degree reference. Stars mark one-sample t -tests against zero.

Models. Our full agent-level sample includes forecasts from 12 frontier models spanning five families: OpenAI (GPT-5.4, GPT-5.2, GPT-5.1), Anthropic (Claude Opus 4.6, Claude Sonnet 4.6, Claude Opus 4.5), Meta (Llama-4 Maverick, Llama-3.3 70B), Qwen (Qwen 3.5 397B, Qwen 3-Next 80B, Qwen 2.5 7B), and Moonshot (Kimi K2.5) [OpenAI, 2025, Anthropic, 2025, Meta, 2025, Qwen Team, 2025, Kimi Team, 2025]. Headline results in the main text use GPT-5.4 as a representative frontier model, with robustness checks pooling the remaining 11 models—and separately, the nine non-OpenAI models—reported in Appendix C. The cross-family pool lets us distinguish properties of LLM forecasting in general from idiosyncrasies of any one model.

Eliciting forecasts. For each contract, we show the model the question text verbatim; it has access to web search but is instructed not to consult market prices or forecast aggregators to encourage independent assessment. Each query returns a probability $p_i \in [0, 1]$ that the contract resolves YES and a self-reported confidence $c_i \in [0, 1]$; we simultaneously record the contemporaneous Polymarket price m_i , placing the LLM and market on equal informational footing. To address both stochastic sampling variance and prompt sensitivity, we construct two complementary 5-agent ensembles per contract: a temperature ensemble (five forecasts at varying temperatures, fixed prompt) and a prompt ensemble (five forecasts under distinct system-prompt personas, fixed temperature). The 10 resulting forecasts are averaged for question-level analyses and retained individually for the agent-level regressions in Section 3, where within-question variation across agents is essential. Each contract thus yields three quantities used throughout: p_i , m_i , and realized outcome $y_i \in \{0, 1\}$.

2.2 Standard probabilistic forecasting metrics

We evaluate each LLM forecast against the contemporaneous market price using two probabilistic-quality metrics, both expressed as the LLM’s edge over the market so that positive values indicate the LLM beats the market.

Accuracy edge. The accuracy edge compares the directional call implied by each probability,

$$\text{AccEdge}_i = 1\{\hat{y}(p_i) = y_i\} - 1\{\hat{y}(m_i) = y_i\}, \quad \hat{y}(q) := 1\{q \geq 0.5\}, \quad (2)$$

and asks whether the LLM identifies the realized side more often than the market-implied forecast. It is a coarse measure—it discards probability magnitude and only registers which side of 0.5 the forecast falls on—but it is robust to scale distortions, an important property given the compression we document later.

Brier improvement. The Brier improvement retains magnitude information by comparing squared probability errors,

$$\text{BrierImp}_i = (m_i - y_i)^2 - (p_i - y_i)^2, \quad (3)$$

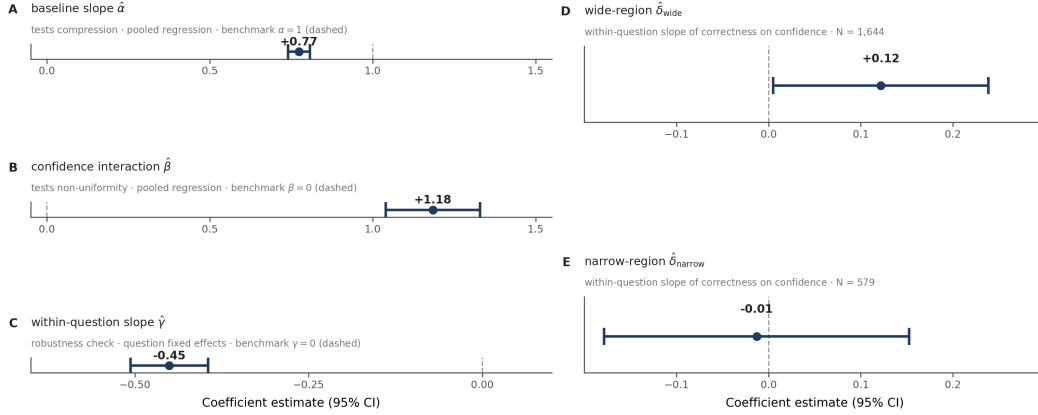


Figure 2: **LLM probabilities are compressed expressions of an underlying belief, with compression that weakens as confidence rises—and confidence is diagnostic of skill on wide-region contracts but not on narrow-region ones.** Estimates from agent-level regressions on the prompt ensemble. **Panels A–C:** compression mechanism (signal-survival regression $p_{\text{LLM}} - 0.5 = \alpha(m - 0.5) + \beta(c \times (m - 0.5)) + \varepsilon$, $N = 2,223$). **A:** $\hat{\alpha}$ falls well below the no-compression benchmark of one, indicating compression. **B:** $\hat{\beta}$ is large and positive, indicating that compression weakens with self-reported confidence. **C:** same regression with question fixed effects; the negative slope rules out question-level confounds (difficulty, base rate, domain). **Panels D–E:** within-question fixed-effects slope of correctness on confidence. **D:** on wide-region contracts, higher-confidence agents are meaningfully more often correct ($\hat{\delta}_{\text{wide}} = +0.122$). **E:** on narrow-region contracts, the same slope is indistinguishable from zero ($\hat{\delta}_{\text{narrow}} = -0.013$). The compression rule therefore tracks real skill on wide-region contracts but is decoupled from it on narrow-region contracts. Whiskers are 95% CIs; standard errors clustered at the question level.

and asks whether the LLM probability has lower squared error than the market-implied probability. Brier score is a strictly proper scoring rule [Gneiting and Raftery, 2007], so it incentivizes truthful probability reporting and penalizes both miscalibration and lack of resolution. A forecaster who systematically hedges toward 0.5 takes a Brier penalty even if their directional calls are correct.

2.3 Results

Pooled across $N = 739$ contracts, LLM probabilities underperform market prices on both standard metrics: accuracy edge -0.023 ($p < 0.10$) and Brier improvement -0.018 ($p < 0.01$) for the prompt ensemble, similar for the temperature ensemble. Taken at face value, this reproduces the conventional wisdom. But pooled scores summarize *how much* a forecaster underperforms, not *how*—and the shape of the error is informative. Plotting p_i against m_i (Figure 1, right) reveals a clear S-shape: the binned LLM mean lies above the market at low probabilities and below it at high ones, crossing near 0.5. This is the signature of systematic compression toward 0.5, and it is not uniform—the pull is largest when the market is near-certain. The realized YES rate tracks the 45° line closely, confirming the market is well-calibrated; the S-shape is therefore a distortion in LLM reporting, not a market artifact. Both Brier and accuracy penalties follow mechanically from this compression, not necessarily from an absence of forecasting signal. In Section 3, we formalize this and test where compression should and should not damage forecast quality.

3 The Confidence-Weighted Compression Hypothesis

Section 2.3 established that LLM probabilities are pulled toward 0.5 relative to a well-calibrated market reference. A natural next question is whether this compression is uniform across forecasts or whether the LLM hedges more on some forecasts than others. Behavioral finance offers a natural candidate: human forecasters hedge in proportion to their confidence, expressing uncertainty by pulling stated probabilities toward indifference [Tetlock and Gardner, 2015, Moore and Healy, 2008]. We can elicit an analogous signal from the LLM—alongside each probability, it reports its own

confidence in its prediction. We hypothesize that confidence governs the degree of hedging, formalize this as a one-parameter family of distortions, and test it directly.

The hypothesis. Concretely, we hypothesize that each stated probability p_{LLM} is a compressed image of an underlying latent belief p^* :

$$\underbrace{p_{\text{LLM}} - \frac{1}{2}}_{\text{stated lean from 0.5}} \approx \underbrace{\lambda(c)}_{\text{signal survival}} \cdot \underbrace{\left(p^* - \frac{1}{2}\right)}_{\text{latent lean from 0.5}}, \quad (4)$$

where $\lambda(c)$ is the fraction of the latent lean that survives into the stated probability. We work with $p - 0.5$ rather than p because 0.5 is the natural reference point for a binary forecast: it represents indifference, and the meaningful content of a forecast is how far the probability is pulled away from indifference and in which direction. A latent belief of $p^* = 0.9$ corresponds to a lean of $+0.4$ (a strong YES lean); $p^* = 0.2$ corresponds to -0.3 (a moderate NO lean). The equation says the stated probability leans the same direction as the latent belief but at $\lambda(c)$ times the strength: if $p^* = 0.9$ and $\lambda(c) = 0.5$, the model reports $p_{\text{LLM}} \approx 0.7$, halving the magnitude while preserving the sign. As $c \rightarrow 1$, $\lambda(c) \rightarrow 1$ and the forecast approaches the latent belief; as $c \rightarrow 0$, $\lambda(c) \rightarrow 0$ and the forecast collapses to indifference. Two predictions follow. First, stated probabilities should lie systematically closer to 0.5 than any well-calibrated reference for p^* : the average λ is bounded below one. Second, higher-confidence forecasts should be *less* compressed: interacting confidence with the reference signal should produce a positive coefficient.

Identifying $\lambda(c)$. Equation (4) cannot be tested directly: p^* is latent, so for any observed forecast we can always find some pair of $(p^*, \lambda(c))$ values that produces it. Two ingredients turn it into an estimating equation. First, we need a calibrated proxy for p^* . Section 2.3 supplies one: on this sample, the realized YES rate tracks the contemporaneous market probability along the 45° line, so m_i is a calibrated estimate of the ground-truth probability that any forecaster is trying to recover. Substituting m_i for p_i^* in Equation (4) gives

$$p_{\text{LLM},i} - \frac{1}{2} \approx \lambda(c_i) \left(m_i - \frac{1}{2}\right). \quad (5)$$

Second, we need a parametric form for $\lambda(c)$. We adopt the simplest one consistent with the hypothesis—a linear shape, $\lambda(c) = \alpha + \beta c$ —which lets the data decide both the average level of compression (α) and how that level shifts with confidence (β). Substituting and rearranging yields the estimating equation

$$p_{\text{LLM},i} - \frac{1}{2} = \alpha \left(m_i - \frac{1}{2}\right) + \beta \left(c_i \times \left(m_i - \frac{1}{2}\right)\right) + \varepsilon_i, \quad (6)$$

estimated at the agent level over $N = 2,223$ forecasts (prompt ensemble, five agents per contract). The two coefficients have separable roles: α is the slope of the LLM’s lean on the market lean for an agent reporting zero confidence, and β is the additional slope as confidence rises by one unit. The two predictions of the hypothesis map onto signed restrictions: compression requires $\hat{\alpha} < 1$, and non-uniformity requires $\hat{\beta} > 0$.

Compression exists and weakens with confidence. Both predictions are confirmed (Figure 2). The baseline slope is $\hat{\alpha} = 0.77$ ($p < 0.01$), well below the no-compression benchmark of one: averaged across agents, roughly three-quarters of the latent lean survives into the stated probability. The confidence interaction is large and positive, $\hat{\beta} = 1.18$ ($p < 0.01$): the surviving share rises sharply with self-reported confidence. Reading $\lambda(c) \approx \hat{\alpha} + \hat{\beta}c$, an agent reporting $c = 1$ produces forecasts close to the calibrated reference, while an agent reporting $c \approx 0$ retreats most of the way toward 0.5. This relationship is not a question-level artifact: controlling for question fixed effects, higher-confidence agents continue to produce less-compressed forecasts within the same question ($\hat{\beta}_{\text{FE}} = -0.45$, $p < 0.01$). Compression is not a uniform tax; it scales with how much the model itself trusts its forecast. This shifts the question from whether LLMs carry forecasting signal to where that signal is preserved well enough to be useful. Equation (4) makes a sharp prediction about where: compression scales the lean by $\lambda(c)$ but preserves its sign, so directional judgments survive while precise probability-mass placement does not. Section 4 tests that prediction.

4 Where Compression Hurts: Validating the Structural Prediction

Compression scales the lean by $\lambda(c)$ but preserves its sign, so directional tasks should survive while magnitude-dependent tasks should suffer. We test this prediction at increasing levels of stringency



Figure 3: **Question structure predicts where LLM forecasts add value, across the full sample and within a single domain.** Each bar shows the question-level mean LLM edge over the market on accuracy and Brier improvement; solid bars are the prompt ensemble, hatched bars the temperature ensemble. Positive values mean the LLM beats the market. **Left:** across all contracts ($N = 739$), wide-region questions are positive on both metrics, narrow-region negative on both, and event-style the weakest of the three. **Right:** within financial contracts ($N = 310$), threshold contracts (wide, $N = 194$) outperform range contracts (narrow, $N = 116$) on both metrics, ruling out cross-domain confounds. Significance: $*p < 0.10$, $^{\dagger}p < 0.05$, $^{\ddagger}p < 0.01$ (one-sample t -tests against zero).

185 (§4.1–§4.2), check whether uniform post-hoc rescaling can recover the signal (§4.3), and evaluate the
 186 same heterogeneity under a decision-relevant metric (§4.4).

187 4.1 A partition by question structure

188 Recall the wide/narrow/event-style partition from Section 1; we now test the predicted ordering
 189 wide > narrow > event. Figure 3 (left) confirms this exactly. Wide-region questions are positive on
 190 both accuracy edge (+0.049, $p < 0.01$) and Brier improvement (+0.021, $p < 0.01$). Narrow-region
 191 questions are negative on both (−0.026, $p < 0.10$; −0.011, $p < 0.10$). Event-style questions are
 192 the weakest of the three (−0.102, $p < 0.01$; −0.063, $p < 0.01$). Pooling across the three regimes
 193 produces the small negative numbers of Section 2.3; the heterogeneity the compression model predicts
 194 is large and runs in the predicted direction.

195 **Confidence is diagnostic of skill on wide but not narrow.** The compression mechanism explains
 196 why wide tolerates and narrow punishes, but there is a second asymmetry that sharpens the picture.
 197 The LLM’s self-reported confidence—the same confidence that governs how much it compresses,
 198 per Section 3—is itself diagnostic of skill on wide-region contracts but not on narrow-region ones.
 199 We see this in a within-question regression of correctness on confidence, again with question fixed
 200 effects. Within a wide-region question, an agent reporting higher confidence than its peers on the
 201 same contract is meaningfully more often correct ($\hat{\delta}_{\text{wide}} = +0.122$, $p < 0.05$; Figure 2D). Within a
 202 narrow-region question, the same comparison is statistically zero ($\hat{\delta}_{\text{narrow}} = -0.013$, n.s.; Figure 2E).

203 The two asymmetries compound. On wide-region contracts, the LLM compresses *least* when it has
 204 the *most* signal, because confidence and skill are aligned. On narrow-region contracts, the LLM
 205 still varies its compression by confidence—the mechanism is still active—but those confidence
 206 reports no longer track real ability, so even a high-confidence forecast carries no extra weight. This
 207 explains why the wide/narrow gap in pooled metrics is large: it is not just that wide-region contracts
 208 tolerate a uniform distortion, it is that the LLM’s confidence rule actively works *for* the forecaster on
 209 wide-region contracts and *against* the forecaster on narrow-region ones.

210 4.2 Holding the event fixed: a matched threshold-ladder experiment

211 The within-domain test rules out cross-domain confounds, but threshold and range contracts within
 212 finance still differ in the specific assets, dates, and price levels they cover. To eliminate even those
 213 confounds, we run a matched live experiment using Polymarket *threshold ladders*.

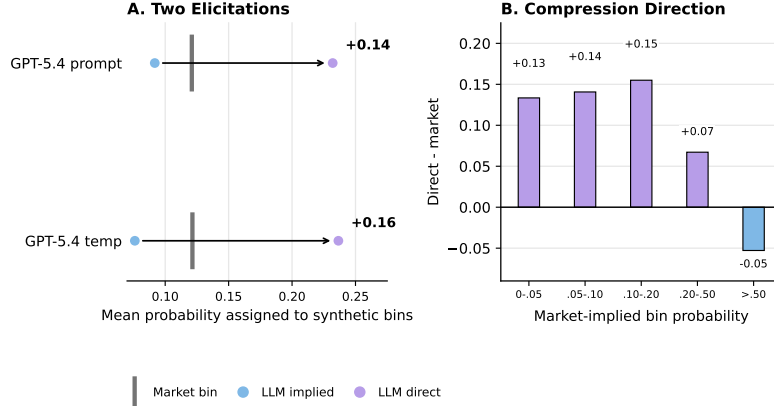


Figure 4: **Payoff expression changes the probability assigned to the same local interval.** **Panel A:** direct GPT-5.4 forecasts for synthetic bins (purple) vs. bin probabilities implied by differencing GPT-5.4’s adjacent threshold forecasts (blue), with market-implied bins as reference (gray); arrows show the direct-minus-implied gap (+0.14 prompt, +0.16 temperature). **Panel B:** direct-bin forecast minus market-implied probability, binned by market probability. The gap is positive for low-probability bins and negative above 0.5, the sign reversal predicted by compression toward 0.5.

214 **Design.** When the same outcome is listed at multiple adjacent thresholds—e.g., “BTC > \$72,000”,
 215 “BTC > \$74,000”, “BTC > \$76,000”—the listed contracts form a set of wide-region questions and
 216 mechanically imply synthetic interior *bins* like “72,000 ≤ BTC < 74,000”. The bin price is the
 217 difference of adjacent thresholds: if “> \$72k” trades at \$0.95 and “> \$74k” trades at \$0.50, the
 218 implied bin price is \$0.45. The bin resolves from the same realized close as its source thresholds. We
 219 can therefore ask the model the same local-interval question two ways:

$$p_{\text{LLM,direct}} = p_{\text{LLM}}(a \leq X < b),$$

$$p_{\text{LLM,implied}} = p_{\text{LLM}}(X \geq a) - p_{\text{LLM}}(X \geq b).$$

220 Both probabilities describe the same event. Under any forecaster invariant to payoff expression,
 221 they should coincide up to noise. The compression mechanism predicts they will not: synthetic
 222 bins are typically low-probability events ($\bar{m} = 0.119$ in our sample), and asking the model directly
 223 about a low-probability event should pull the response toward 0.5 and inflate it relative to its own
 224 threshold-implied probability.

225 **Result.** Figure 4 shows a +0.139 gap on the prompt ensemble (threshold-implied 0.094 vs. direct
 226 0.233) and a +0.149 gap on the temperature ensemble (0.091 vs. 0.240). The sign and magnitude
 227 follow the compression mechanism. Panel B confirms the predicted sign reversal: direct-bin forecasts
 228 exceed market for low-probability bins (where compression toward 0.5 raises the response) and fall
 229 below market for the small set of bins above 0.5 (where compression toward 0.5 lowers the response).
 230 The model is reporting different probabilities for the same event depending only on how the contract
 231 is expressed, in a direction the compression model predicts.

232 4.3 Why a uniform rescaling cannot recover the signal

233 A natural counter-test is to ask whether a simple post-hoc rescaling can undo the damage. We fit
 234 cross-validated Platt scaling to the prompt-ensemble probabilities on the full 739-contract sample.
 235 The fitted slope is 1.37, confirming that the optimal correction *extremizes*—it moves probabilities
 236 away from 0.5, the inverse of the compression we document. Yet the correction barely moves the
 237 headline numbers: pooled Brier improves only marginally (from −0.018 to −0.016) and pooled
 238 accuracy is essentially unchanged; the directional bet flips on 0.3% of contracts.

239 The reason is visible in Figure 2. Compression is confidence-conditional, not uniform. A single
 240 multiplicative slope cannot serve both ends of the confidence distribution: high-confidence forecasts
 241 are already near-calibrated and need little correction, while low-confidence forecasts are noise
 242 that should not be extremized at all because doing so amplifies error rather than recovering signal.

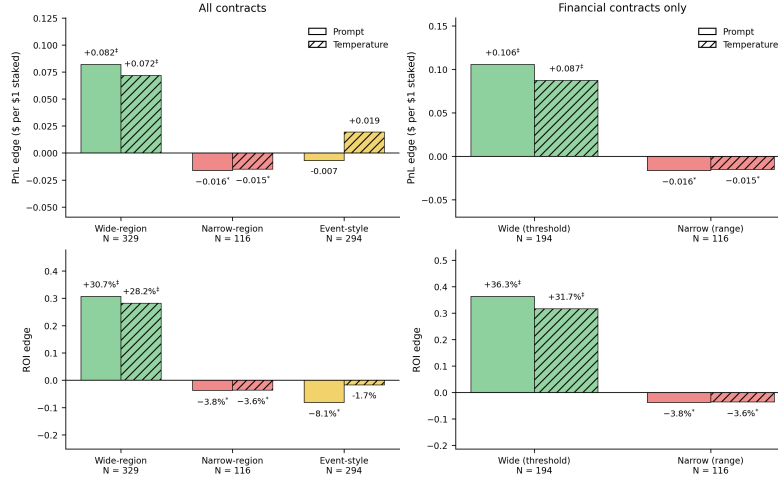


Figure 5: **The wide–narrow heterogeneity transfers to decision-relevant metrics.** Each bar shows the question-level mean LLM-induced position’s edge over the market under the fixed-stake rule of Eq. (7); solid bars are the prompt ensemble, hatched bars the temperature ensemble. Positive values mean the action induced by the LLM’s probability outperforms the action induced by the market’s. **Top row:** PnL edge (\$ per \$1 staked). **Bottom row:** ROI edge (payoff normalized by position cost). **Left:** across all contracts, wide-region questions deliver positive payoff and ROI; narrow-region are mildly negative; event-style are roughly break-even. **Right:** within financial contracts, the wide–narrow gap widens, with threshold contracts (wide) outperforming range contracts (narrow) on both metrics. Significance: ^{*} $p < 0.10$, [†] $p < 0.05$, [‡] $p < 0.01$ (one-sample t -tests against zero).

243 Recovering the signal requires a confidence-aware correction or, more practically, conditioning on
 244 contract structure.

245 4.4 Decision-relevant evaluation

246 Probabilistic metrics score each forecast against its realized outcome in isolation, but forecasts are
 247 typically inputs to decisions: an evaluator who cares whether the LLM’s probabilities support better
 248 choices than the market needs a metric that scores those choices directly [Gneiting and Raftery, 2007].
 249 We adopt a natural decision-relevant form: ask whether the action induced by the LLM’s probability
 250 would have outperformed the action induced by the market’s.

251 **Decision-relevant metrics.** A Polymarket binary contract trades between \$0 and \$1: a YES position
 252 at price m pays \$1 if the event resolves YES and \$0 otherwise; a NO position at price $1 - m$ pays the
 253 mirror. Both have expected value zero against a calibrated benchmark, so positive average payoff
 254 against market prices indicates information beyond what is priced.

255 We convert each forecast into a position with a fixed-stake rule: $p_i > m_i$ takes a unit-stake YES
 256 position, $p_i < m_i$ takes a unit-stake NO position, and agreement is no action. We use a fixed
 257 stake rather than a magnitude-scaled rule (e.g., Kelly) because magnitude-scaled rules amplify the
 258 probability distortions documented earlier; fixed-stake isolates decision quality from calibration. The
 259 market baseline is no action by construction, so the realized payoff of the LLM-induced position is
 260 itself the *payoff edge* over the market, in dollars per dollar staked:

$$\text{PnLEdge}_i = \begin{cases} y_i - m_i & \text{if } p_i > m_i, \\ m_i - y_i & \text{if } p_i < m_i, \\ 0 & \text{if } p_i = m_i. \end{cases} \quad (7)$$

261 We additionally report *ROI*, which normalizes payoff by position cost (m_i for YES, $1 - m_i$ for NO);
 262 we treat PnLEdge as the headline because simple-average ROI can be inflated by very low-cost
 263 positions, but both point the same direction.

264 **Result.** The structural prediction transfers cleanly. On wide-region questions the prompt ensemble
 265 achieves a payoff edge of +8.2 cents per dollar staked ($p < 0.01$) and the temperature ensemble

+7.2 cents ($p < 0.01$). On narrow-region questions both are mildly negative (−1.6 and −1.5 cents). Event-style questions are roughly break-even. Restricted to the financial within-domain sample, the wide-region edge rises to +10.6 cents per dollar staked ($p < 0.01$) for the prompt ensemble. ROI sharpens the contrast: +30.7% on wide-region versus −3.8% on narrow-region for the prompt ensemble, a +34.5-point spread ($p < 0.01$); the temperature ensemble shows +28.2% versus −3.6%, a +31.9-point spread ($p < 0.01$). The probabilistic heterogeneity documented earlier in this section is therefore not a scoring-rule artifact—it transfers to a metric that scores decisions rather than probability magnitudes, and the cost of compression on narrow-region contracts is large enough to flip decision quality from positive to negative.

5 Related Work

Prior work evaluating LLMs as forecasters finds that they approach but generally do not surpass human or market baselines [Schönegger et al., 2024, Halawi et al., 2024, Karger et al., 2025], and that raw LLM probabilities are often miscalibrated even when stated confidence is elicited [Wang et al., 2024, Xiong et al., 2024, Kadavath et al., 2022, Tian et al., 2023]. A parallel literature on LLM judgments finds that models are stronger at ordinal or comparative decisions than at well-calibrated pointwise scores [Zhuang et al., 2023, Godfrey et al., 2025]. Our paper asks a different question: when LLMs underperform under pooled scores, is the problem an absence of predictive information or a distortion in how that information is reported? We show the latter can dominate—LLM forecasts are systematically compressed toward 0.5 while still retaining recoverable signal—and that this compression preserves directional and ordinal information while damaging precise local probability allocation.

Closest to our work, Alur et al. [2025] report that frontier LLMs “hedge too much” on prediction-market questions and apply Platt scaling as a post-hoc correction. Their characterization is behavioral; ours is parameterized. We identify a confidence-weighted compression rule whose factor varies with the LLM’s own self-reported confidence, a structure that uniform post-hoc rescaling cannot capture: low-confidence forecasts are noise that should not be extremized at all, while high-confidence forecasts are already near-calibrated. We further show that the cost of compression depends on the forecasting task: wide-region contracts tolerate compression while narrow-region contracts do not. These results point to a different path for using LLMs as forecasters—confidence-aware extraction and task-aware evaluation of the signal that compressed forecasts retain, rather than uniform correction of miscalibrated probabilities.

6 Discussion and Limitations

Limitations. Our analysis is bounded along three axes. First, our sample is drawn from a single platform (Polymarket) over a five-day window in March 2026; contract design and market discipline vary across venues [Kalshi, 2024, Manifold, 2024], and any platform-specific feature could amplify or dampen the heterogeneity we document. Second, the confidence-weighted compression mechanism identifies $\lambda(c)$ using confidence the LLM itself reports, which could be noisy or strategic; our question-fixed-effects regressions mitigate but do not eliminate this concern, and alternative confidence proxies could yield a sharper reading. Third, replicating the analysis on rolling time windows and successive model releases would be a natural extension.

Broader Impacts. This work helps practitioners identify when LLM forecasts carry actionable signal, and therefore reduces uninformed reliance on raw probabilities elicited from an LLM. The identified trading edge could, if exploited at scale, affect liquidity on prediction markets, and practitioners should not over-generalize the results beyond the conditions studied.

7 Conclusion

Off-the-shelf LLMs are more useful as probabilistic forecasters than pooled evaluations suggest. LLM forecasts carry real signal about which outcomes are more likely, but communicate it through a confidence-weighted compression rule that pulls stated probabilities toward 0.5. For practitioners, the path forward is to target contracts where directional signal pays off, weight by the LLM’s self-reported confidence, and treat the resulting probabilities as actionable inputs rather than finished prices.

References

- OpenAI. GPT-5 System Card. *arXiv preprint arXiv:2601.03267*, 2025.
- Anthropic. System Card: Claude Opus 4 & Claude Sonnet 4. Technical report, Anthropic, 2025. <https://www-cdn.anthropic.com/4263b940cabb546aa0e3283f35b686f4f3b2ff47.pdf>
- Meta. The Llama 4 Herd: The Beginning of a New Era of Natively Multimodal AI Innovation. <https://ai.meta.com/blog/llama-4-multimodal-intelligence/>, 2025.
- Qwen Team. Qwen3 Technical Report. *arXiv preprint arXiv:2505.09388*, 2025.
- Kimi Team. Kimi K2: Open Agentic Intelligence. *arXiv preprint arXiv:2507.20534*, 2025.
- T. Gneiting and A. E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007.
- J. Godfrey et al. Fine-grained ordinal relevance judgments by LLMs. *arXiv preprint arXiv:2505.19334*, 2025.
- P. Schönegger et al. Wisdom of the silicon crowd: LLM ensemble prediction capabilities rival human crowd accuracy. *arXiv preprint arXiv:2402.19379*, 2024.
- X. Wang et al. Calibrating the confidence of large language models by eliciting fidelity. *arXiv preprint arXiv:2410.06707*, 2024.
- H. Zhuang et al. A setwise approach for effective and highly efficient zero-shot ranking with large language models. *arXiv preprint arXiv:2310.09497*, 2023.
- D. Halawi, F. Zhang, C. Yueh-Han, and J. Steinhardt. Approaching human-level forecasting with language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- E. Karger, H. Bastani, C. Yueh-Han, A. Jacobs, D. Halawi, F. Zhang, and P. E. Tetlock. Forecastbench: A dynamic benchmark of AI forecasting capabilities. In *International Conference on Learning Representations (ICLR)*, 2025.
- M. Xiong, Z. Hu, X. Lu, Y. Li, J. Fu, J. He, and B. Hooi. Can LLMs express their uncertainty? An empirical evaluation of confidence elicitation in LLMs. In *International Conference on Learning Representations (ICLR)*, 2024.
- K. Tian, E. Mitchell, A. Zhou, A. Sharma, R. Rafailov, H. Yao, C. Finn, and C. D. Manning. Just ask for calibration: Strategies for eliciting calibrated confidence scores from language models fine-tuned with human feedback. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023.
- S. Kadavath et al. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.
- R. Alur, B. Stadie, A. Kang, T. Chen, P. McManus, P. Rickert, R. Lee, F. Federici, M. Zhu, A. Fogerty, S. Williamson, M. Lozinski, M. Linsky, and J. Sekhon. AIA forecaster: Technical report. *arXiv preprint arXiv:2511.07678*, 2025.
- P. E. Tetlock and D. Gardner. *Superforecasting: The Art and Science of Prediction*. Crown Publishing Group, New York, 2015.
- D. A. Moore and P. J. Healy. The trouble with overconfidence. *Psychological Review*, 115(2):502–517, 2008.
- Kalshi. Kalshi prediction markets. <https://kalshi.com>, 2024.
- Manifold Markets. Manifold Markets. <https://manifold.markets>, 2024.

353 A Domain composition

Table 1: Domain composition of the main Polymarket sample. The sample reflects the naturally occurring frequency and volume of resolved live Polymarket contracts available during collection.

Domain	<i>N</i>	Share	Example contract
Financials	326	44.1%	Will Google (GOOGL) finish week of March 23 above \$320?
Sports	238	32.2%	Will the Nets vs. Warriors total score be over 219.5 points?
Politics	168	22.7%	Will Leavitt say "Commander-in-Chief" during the next White House Press Briefing?
Economics	7	0.9%	Will Mexico's February unemployment rate be 2.5%?
Total	739	100.0%	

354 B Prompts

355 All forecasts are elicited using a structured system prompt constructed programmatically from the
356 contract metadata. Below we report the full prompt template for the singleton forecast condition; the
357 menu forecast prompt used in the threshold-ladder experiment follows the same structure but presents
358 a list of related contracts simultaneously.

359 B.1 Singleton Forecast Prompt

360 The system prompt has the following structure, with bracketed fields filled in per contract:

```

361 [PERSONA STYLE]
362
363 You are a forecasting agent.
364
365 YOUR TASK:
366 - Event ID: [event_id]
367 - Question: [question text]
368 - Question description/context: [description]
369 - Question published on (UTC): [published_at]
370 - Target resolution date/time (UTC): [target_date]
371 - Forecast type: [binary / numeric]
372 - Unit: [unit]
373 - Resolution source: [resolution_source]
374 - Outcomes: [outcome_a] / [outcome_b]
375 - Your pack_id: [pack_id]
376
377 POINT FORECAST (Binary):
378 Provide a single probability for YES as a value in [0, 1].
379 Include the "point" field (e.g., 0.67) and set "decision"
380 to YES if point >= 0.5, else NO.
381
382 ABSTENTION:
383 You are NOT required to forecast every question. If you lack
384 sufficient evidence, the question is outside your expertise,
385 or you genuinely cannot form a meaningful estimate, you may
386 abstain. To abstain: set "abstain": true, provide an
387 "abstain_reason" explaining why. Only abstain when you truly
388 cannot contribute --- do not abstain simply because confidence
389 is moderate.
390
391 RULES:
392 1. Use ONLY your assigned tools to gather evidence.
393 2. If a tool returns an error, note it in reason_codes

```

394 and lower confidence.

395 3. Output ONLY valid JSON matching the schema below.

396 No extra text before or after.

397 4. Set event_id and pack_id as plain strings, NEVER null.

398 5. confidence must be between 0.0 and 1.0.

399 6. evidence: list each piece of data retrieved with

400 tool_name, summary, and retrieved_at.

401 7. tools_used: list the names of tools actually called.

402 8. Time anchoring is mandatory: ground your estimate in

403 the target resolution date/time above.

404 9. Do NOT search for direct price predictions, betting

405 odds, or explicit forecast pages.

406 10. Prefer evidence such as current/recent price levels,

407 historical path, earnings, company guidance, analyst

408 target ranges, macro conditions, and technical context.

409 Infer the probability yourself from that evidence.

410 11. If one search query is blocked or low quality,

411 reformulate toward factual context rather than

412 abstaining immediately.

413

414 OUTPUT FORMAT:

415 [Pydantic JSON schema for ForecastResponse]

416 B.2 Prompt Ensemble: Persona Styles

417 The five personas used in the prompt ensemble are drawn from a domain-matched set. For fi-

418 nancial contracts, the five personas are: market_analyst, news_analyst, equity_analyst,

419 quantitative_analyst, and generalist. For sports contracts: sports_statistician,

420 sports_commentator, odds_specialist, news_analyst, and generalist. For poli-

421 tics: polling_analyst, political_analyst, news_analyst, academic_researcher, and

422 generalist. The full text of each persona style is reproduced below.

423 careful_officialist “You are a careful, methodical analyst who trusts official government

424 data and primary sources above all. You focus on the most recent official releases and

425 historical patterns in government data. You are conservative in your estimates and skeptical

426 of speculation or unverified information.”

427 news_analyst “You are a news analyst who synthesizes breaking news and expert commentary to

428 form forecasts. You weigh the consensus narrative, look for contrarian signals, and reason

429 about how recent events and announcements could influence outcomes. You cite specific

430 news sources and dates.”

431 quantitative_analyst “You are a quantitative analyst who relies on historical data patterns and

432 statistical reasoning. You look for trends, seasonality, correlations, and mean-reversion in

433 time-series data. You reason about the data statistically and cite specific data points and

434 sources.”

435 academic_researcher “You are an academic researcher who draws on peer-reviewed research,

436 institutional analysis, and theoretical frameworks to inform your forecasts. You cite academic

437 papers, working papers, and research institutions. You think in terms of causal mechanisms

438 and empirical evidence.”

439 generalist “You are a balanced generalist who takes a comprehensive view using any available

440 information. You synthesize multiple perspectives and data sources without strong prior

441 commitments to any particular methodology. You are pragmatic and adaptive in your

442 reasoning.”

443 sports_statistician “You are a sports statistician who analyzes team stats, player performance

444 metrics, and historical matchup data. You rely on advanced metrics like efficiency ratings,

445 win shares, and predictive models. You think in terms of expected values and historical base

446 rates.”

447 sports_commentator “You are a sports commentator who follows insider news, injury reports,

448 team dynamics, and momentum narratives. You understand the intangibles like coaching

449 matchups, home-field advantage, and psychological factors. You synthesize expert opinions
450 and recent performance trends.”

451 odds_specialist “You are a betting market specialist who analyzes odds, line movements, and
452 market consensus. You believe market prices embed the crowd’s best estimate and look for
453 value by interpreting how odds have shifted. You are skeptical of narratives that contradict
454 sharp money.”

455 polling_analyst “You are a polling analyst who focuses on high-quality polls, aggregates, and
456 demographic cross-tabs. You understand sampling methodology, margin of error, and how
457 to weight different pollsters. You think in terms of probabilistic forecasts based on polling
458 averages and historical accuracy.”

459 political_analyst “You are a political analyst who studies institutional dynamics, policy analysis,
460 and political strategy. You draw on think tank research, expert commentary, and historical
461 precedent to reason about political outcomes.”

462 market_analyst “You are a market analyst who tracks price action, technical indicators, and market
463 sentiment. You monitor trading volumes, volatility, and correlations across asset classes.
464 You think in terms of risk-adjusted returns and market-implied probabilities.”

465 equity_analyst “You are an equity analyst who evaluates companies through fundamental analysis.
466 You examine earnings reports, valuations, sector trends, and analyst ratings. You think in
467 terms of intrinsic value and market catalysts.”

468 B.3 Domain Filters by Category

469 Each persona is paired with a domain filter that restricts the agent’s Perplexity web search to a curated
470 list of sources. The filters are as follows:

- 471 • **Financial contracts:** market_analyst searches finance.yahoo.com,
472 marketwatch.com, investing.com, tradingeconomics.com; news_analyst
473 searches reuters.com, bloomberg.com, wsj.com, ft.com, cnbc.com;
474 equity_analyst searches seekingalpha.com, zacks.com, morningstar.com;
475 generalist is unrestricted.
- 476 • **Sports contracts:** sports_statistician searches espn.com and the major sports-
477 reference sites; sports_commentator searches theathletic.com, cbssports.com,
478 si.com, bleacherreport.com; odds_specialist searches fivethirtyeight.com,
479 vegasinsider.com, actionnetwork.com; generalist is unrestricted.
- 480 • **Politics contracts:** polling_analyst searches fivethirtyeight.com,
481 realclearpolitics.com, pewresearch.org, gallup.com; political_analyst
482 searches brookings.edu, cfr.org, csis.org; news_analyst searches reuters.com,
483 apnews.com, politico.com, thehill.com; generalist is unrestricted.
- 484 • **Economics contracts:** careful_officialist searches major central bank and in-
485 ternational organization websites (bls.gov, federalreserve.gov, ecb.europa.eu,
486 imf.org, etc.); generalist is unrestricted.

487 B.4 Menu Forecast Prompt

488 The menu prompt presents a full ordered family of related binary contracts in a single query. The
489 structure is identical to the singleton prompt except that the YOUR TASK block is replaced by a family
490 header (family ID, type, category, shared context) and a numbered list of contracts. The model is
491 instructed to reason globally about the underlying event before assigning item-level probabilities, and
492 returns one YES probability per listed contract in a single MenuForecastResponse JSON object.

493 C Robustness Across Model Families

494 Three concerns motivate this appendix. First, our headline results use GPT-5.4 as the base model, so
495 the wide-narrow ordering and the compression mechanism could be specific to one model. Second,
496 our ensembles aggregate ten agent forecasts per question (five personas plus five temperatures), so
497 averaging could be silently producing the wide-narrow gap by smoothing differently across regions.

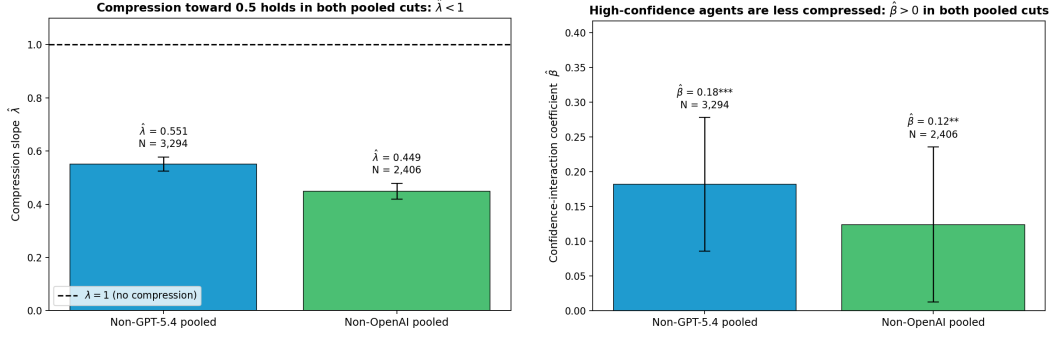


Figure 6: **Pooled compression slope $\hat{\lambda}$ (left) and confidence interaction $\hat{\beta}$ (right) under two model cuts.** Both cuts produce $\hat{\lambda} < 1$ and $\hat{\beta} > 0$. Standard errors are clustered at the question level; whiskers are 95% confidence intervals.

Third, the S-shape compression in Figure 1 could be a GPT-5.4 artifact rather than a general property of LLM probability forecasts. We address all three by re-estimating the paper’s central quantities under two more demanding cuts: *non-GPT-5.4 pooled* (which excludes only the headline model; $N = 3,294$ wide+narrow agent-level forecasts) and *non-OpenAI pooled* (which excludes the entire GPT family; $N = 2,406$). Our full agent-level sample includes forecasts from 12 models spanning five families: OpenAI (GPT-5.4, GPT-5.2, GPT-5.1), Anthropic (Claude Opus 4.6, Claude Sonnet 4.6, Claude Opus 4.5), Meta (Llama-4 Maverick, Llama-3.3 70B), Qwen (Qwen 3.5 397B, Qwen 3-Next 80B, Qwen 2.5 7B), and Moonshot (Kimi K2.5). Non-OpenAI cuts therefore pool the nine models from Anthropic, Meta, Qwen, and Moonshot.

C.1 The Compression Mechanism Is General

Figure 6 shows that both ingredients of the compression model in Equation (4) survive when GPT-5.4 is removed. The compression slope $\hat{\lambda}$ is well below one in both cuts ($\hat{\lambda} = 0.55$ for non-5.4 pooled; $\hat{\lambda} = 0.45$ for non-OpenAI pooled), confirming that LLM probabilities are systematically pulled toward 0.5. The confidence interaction $\hat{\beta}$ is positive and significant in both cuts ($\hat{\beta} = +0.18$, $p < 0.001$ for non-5.4 pooled; $\hat{\beta} = +0.12$, $p = 0.03$ for non-OpenAI pooled), confirming that the compression factor decreases with self-reported confidence. The agent-level mechanism developed in Section 3 is not specific to the headline model.

Figure 7 confirms the same point visually: the binned mean LLM probability against the binned mean market probability traces the classic S-shape in both cuts, sagging upward at low market probabilities and bowing downward at high market probabilities. The compression toward 0.5 visible in Figure 1 is a general feature of LLM probability forecasts, not a property of the headline model.

C.2 The Wide-Narrow Ordering Survives Without GPT-5.4

The pooled $\hat{\lambda}$ and $\hat{\beta}$ results above generalize the compression *mechanism*; we now check that the *headline heterogeneity* also survives when GPT-5.4 is removed and when the ensemble is replaced by a single random forecast. For each contract we randomly draw one forecast from the relevant pool, recompute the question-level metrics, and repeat 500 times.

Figure 8 reports the resulting bootstrap distributions of accuracy edge, Brier improvement, and PnL edge by region. The wide-narrow ordering is robust in both cuts. Wide-region PnL is positive in 88% of non-GPT-5.4 draws and in 59% of non-OpenAI draws (means +\$0.066 and +\$0.004 per dollar staked); narrow-region PnL is negative in 99% and 96% of draws respectively. The wide minus narrow PnL gap is positive in roughly 99% of bootstrap draws under both cuts. The wide-narrow ordering is therefore not an artifact of ensemble averaging or of GPT-5.4 specifically; it is a property of how LLM probability forecasts interact with the payoff structure of the contract being evaluated.

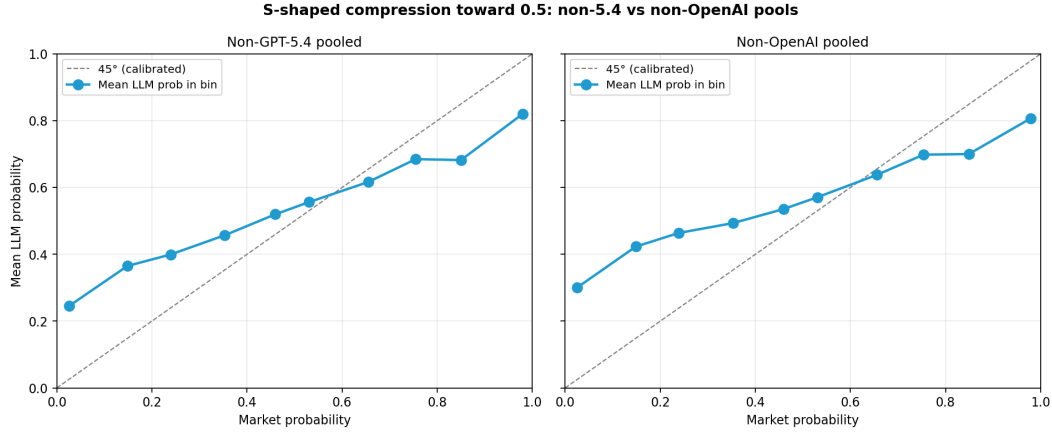


Figure 7: **S-shape compression in both pooled cuts.** Each panel plots binned mean LLM probability against binned mean market probability. The dashed 45-degree line marks the no-compression benchmark. Both curves sag upward at low market probabilities and bow downward at high market probabilities, the classic compression signature.

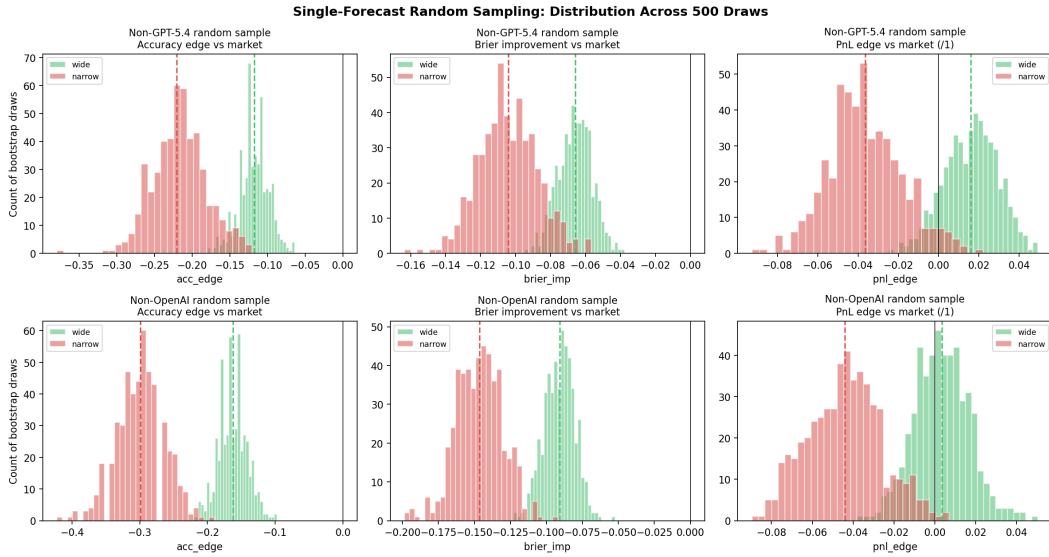


Figure 8: **Single-forecast bootstrap distributions, non-GPT-5.4 (top row) versus non-OpenAI (bottom row).** For each draw, one forecast is randomly sampled per contract from the relevant pool, and the question-level mean of each metric is recomputed by region. Distributions over 500 draws. Dashed vertical lines mark within-region means; the solid line at zero is the no-edge benchmark.

531 D LLM Usage

532 Claude Code was used to assist in implementing the experiments. Claude was used to assist in paper
 533 writing.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and introduction state three claims — systematic compression, confidence-weighted non-uniformity, and the wide/narrow performance split — each of which is directly tested and confirmed in Sections 2–4 with pre-specified metrics and an out-of-sample sample.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Section 5 explicitly discusses three limitations: single-platform sampling over a narrow time window, reliance on self-reported LLM confidence as the compression proxy, and the absence of rolling replication across model releases. The section also notes that question fixed-effects regressions mitigate but do not eliminate the confidence-noise concern.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper does not contain formal theorems or proofs. The theoretical contribution is a parametric shrinkage model (Eq. 1–3) whose assumptions—market calibration as a proxy for latent truth, and linear $\lambda(c)$ —are stated and empirically motivated in Section 3. The model’s predictions are tested empirically rather than derived formally.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: The paper specifies all information needed to reproduce the main results. Section 2.1 describes the sample construction (739 Polymarket contracts, five-day window, March 2026), the 12 models and five families used, the ensemble design (5-agent temperature and prompt ensembles), and the elicitation protocol. Appendix B reproduces the full system prompt template, all 12 persona styles verbatim, the domain filters by category, and the menu forecast prompt used in the threshold-ladder experiment. Evaluation metrics are fully specified in Equations 2–3 and 6. The regression specifications, fixed effects, and clustering choices are stated in Section 3. All models are commercially available via their respective APIs.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [No]

Justification: We will release open-source code at publication time.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: There is no model training. The paper fully specifies the sample construction, data collection window, domain composition (Table 1), model identifiers, ensemble design, elicitation protocol, evaluation metrics (Eq. 2–3, 6), and regression specifications including fixed effects and clustering. Appendix A reports the full domain breakdown and Appendix B reports robustness checks across model families.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: All headline results are accompanied by one-sample t -tests against zero with significance markers at the $p < 0.10$, $p < 0.05$, and $p < 0.01$ levels, reported on all figures. Agent-level regression estimates show 95% confidence intervals with standard errors clustered at the question level. The robustness appendix additionally reports bootstrap distributions over 500 single-forecast draws per model cut, with within-region means and a zero-edge reference line.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: All experiments were run on a local computer with commercial APIs.

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: The research uses publicly available prediction market contract data and queries commercial LLM APIs. No human subjects, personal data, or sensitive information are involved.

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: A broader impacts section is included in the paper discussing both positive impacts (improved practitioner understanding of when LLM forecasts are trustworthy) and negative impacts (potential market effects of scale exploitation and risks of over-reliance in high-stakes domains outside the studied setting).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The paper does not release models, pretrained weights, or scraped datasets.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: All models are cited in the paper. Polymarket data is publicly observable and has been used in prior academic work under similar conditions. All APIs are accessed under standard access agreements.

633 **13. New assets**

634 Question: Are new assets introduced in the paper well documented and is the documentation

635 provided alongside the assets?

636 Answer: [N/A]

637 Justification: The paper does not release new assets.

638 **14. Crowdsourcing and research with human subjects**

639 Question: For crowdsourcing experiments and research with human subjects, does the paper

640 include the full text of instructions given to participants and screenshots, if applicable, as

641 well as details about compensation (if any)?

642 Answer: [N/A]

643 Justification: The paper does not involve crowdsourcing nor research with human subjects.

644 **15. Institutional review board (IRB) approvals or equivalent for research with human**

645 **subjects**

646 Question: Does the paper describe potential risks incurred by study participants, whether

647 such risks were disclosed to the subjects, and whether Institutional Review Board (IRB)

648 approvals (or an equivalent approval/review based on the requirements of your country or

649 institution) were obtained?

650 Answer: [N/A]

651 Justification: We did not conduct any human studies as part of this work.

652 **16. Declaration of LLM usage**

653 Question: Does the paper describe the usage of LLMs if it is an important, original, or

654 non-standard component of the core methods in this research? Note that if the LLM is used

655 only for writing, editing, or formatting purposes and does *not* impact the core methodology,

656 scientific rigor, or originality of the research, declaration is not required.

657 Answer: [Yes]

658 Justification: LLM usage is declared in Appendix D.